MATLAB Program

```
% PID Controller Tuning and Response Analysis

clc;
clear;
close all;
s = tf('s');
% Plant Transfer Function
G = 1/(s^2 + 10*s + 20);
% PID Parameters
Kp = 30;
Ki = 70;
Kd = 2;

C = pid(Kp,Ki,Kd);
% Closed Loop System
T = feedback(C*G,1);

% Step Response
figure;
step(T);
grid on;
title('PID Controlled System Response');
% Performance Analysis
info = stepinfo(T)
disp('Rise Time = ');
```

```
disp(info.RiseTime);
disp('Settling Time = ');
disp(info.SettlingTime);

disp('Overshoot = ');
disp(info.Overshoot);
```

Simulink

